# Junior Deogracias MAMBO MOTSOU

## FPGA & SoC Engineer — VHDL, Vivado, Vitis

motsoujunior17@gmail.com | +33 6 01 16 86 06 | France | Available now
teo1747.github.io/Portfolio | github.com/teo1747 | linkedin.com/in/teo1747

## Profile

Embedded systems engineer (Polytech Dijon, 2026) focused on digital design for FPGAs and Zynq SoCs: RTL in VHDL, AXI4-Lite and AXI4-Stream interfaces, Vivado IP integration and block design, and the bare-metal software that drives the hardware from Vitis. I verify RTL against bit-exact C models with self-checking testbenches. A year of automotive firmware bring-up at FORVIA gives me the system view.

## Technical skills

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|---|---|
| **HDL & design** | VHDL (synthesisable RTL, VHDL-2008 testbenches), Verilog (basic); FSMs, pipelined datapaths, line buffers, bit-growth management, AXI handshakes with back-pressure, bidirectional I/O |
| **AMD/Xilinx flow** | Vivado: IP integrator block design, IP packaging, synthesis, implementation, XDC, timing and utilisation reports, bitstream, ILA and Hardware Manager; Vitis HLS; Vitis bare-metal applications; TCL scripting of the flow |
| **Devices & boards** | Zynq-7000 (ZedBoard, xc7z020), Artix-7 (Nexys 4 DDR, xc7a100t) |
| **Interfaces & IP** | AXI4-Lite, AXI4-Stream, AXI DMA, Zynq PS7 (GP/HP ports), Video Timing Controller, Clocking Wizard, FIFO Generator; VGA, I²C, UART, SPI, ZBT SRAM |
| **Verification** | Self-checking testbenches, bit-exact C reference models, random stall injection; GHDL, XSim, ModelSim; CI simulation on GitHub Actions |
| **Embedded software** | C (bare-metal, Xilinx standalone drivers, CMSIS), FreeRTOS, Zephyr; Python, TCL |
| **Debug** | ILA, JTAG / Hardware Manager, J-Link, GDB, serial consoles |

## Professional experience

### Embedded Systems Engineer — apprentice (contrat de professionnalisation)

Sep. 2025 – Sep. 2026

FORVIA – Faurecia Interior Systems Surface Activation department, France

Firmware bring-up: LIN-controlled heating-element controller (TCU)

- Developed C firmware on the Infineon TLD4020-3ET (Arm Cortex-M23 with integrated LIN): LIN ON/OFF command handling, heater PWM through the LDRV peripheral, NTC-based temperature regulation, fault protection and a state-machine architecture.
- Diagnosed early bring-up blockers: traced a HardFault in device initialisation to a silicon-revision mismatch on the evaluation board (register absent on older silicon), and a lost debug connection to boot-mode jumper and USB power conditions.
- The NTC divider is supplied from the switched heater output, so the ADC node is a 100 Hz square wave during PWM. Synchronised sampling to the PWM on-time (ISR at mid-pulse, 16-sample averaging), which brought measurement scatter within the regulation requirement.

Environment: TLD4020-3STD_EVAL board, IAR Embedded Workbench, Infineon MCU Configuration Wizard, on-board J-Link, printf over LIN (MobaXterm, 19 200 baud 8N1)

MeLiBu LED driver port: Melexis MLX81116 to MLX81143

- Ported an existing MeLiBu 1.x driver on RP2040 to the MLX81143: extended-mode sub-address encoding (14 LEDs per node), linear 15-bit PWM encoding confirmed against the datasheet and on the hardware, for a 22 × 7 panel (154 RGB LEDs).
- Fixed bus-timing faults: break generation with busy-wait while interrupts are masked, critical section narrowed to the break to stop UART RX byte loss, TX-FIFO drain to prevent back-to-back frame collisions; corrected the bus return pin.
- Isolated the panel's LED wiring scramble in the hardware-mapping layer, separate from the input convention; used a PIO-based receiver to confirm slave acknowledgements.

Environment: RP2040, Raspberry Pi Pico SDK, PIO state machines, C, MeLiBu 1.x

Interior ambient-lighting demonstrator (software owner)

- Extended the control software of an automotive interior demonstration buck from a few LED displays to nine STM32 UART gateways driving WS2812 LED strips across several lighting zones.
- Wrote a multithreaded Python/PyQt5 application: OpenCV decodes MP4 animations, maps logical LED IDs to frame pixels and distributes each frame to one worker thread per gateway at 921 600 baud, with a barrier to keep all nine gateways in step; custom serial frames (ASCII with delta compression).
- Built an Android tablet app (Kotlin, Jetpack Compose) that controls zones, colours and brightness presets over Bluetooth serial using a JSON command protocol with explicit error codes; added a keyboard fallback mode.
- Adapted the STM32 gateway firmware with minimal changes (ID-offset filtering when two gateways share a UART line).

Environment: Python 3, PyQt5, OpenCV, Kotlin/Jetpack Compose, STM32 (C), WS2812, UART, Bluetooth SPP, Git

**Embedded Systems Intern** Apr. – Aug. 2025
KMUTT — MDT Lab Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.
- FPGA ramp-up: designed an 8-bit microcontroller in VHDL (12-state fetch–decode–execute controller, ALU with NZVC flags, memory-mapped ROM/RAM/I/O) following the LaMeres reference architecture; simulated in Vivado 2024.2 / XSim.

Environment: Python, TensorFlow/Keras, OpenCV, VHDL, Vivado 2024.2

## FPGA & SoC projects

### Personal

Personal projects. I set the architecture and verification plan; implementation is assisted by AI coding tools. Verified in simulation and emulation, not yet on physical boards.

**Image-filter accelerator for Zynq-7000 with AXI DMA** personal Oct. 2026
github.com/teo1747/zynq-image-filter-accelerator

- Streaming 3×3 filter IP in VHDL (box, Gaussian, Sobel), one pixel per clock: line buffers, 4-stage pipeline, full AXI4-Stream back-pressure, TLAST/TUSER framing; AXI4-Lite control and status registers; packaged for Vivado IP integrator.
- Zynq PS7 + AXI DMA block design scripted in TCL through bitstream and XSA; bare-metal Vitis application streams frames from DDR, checks every pixel against the C model and times HW against SW.
- Bit-exact against a C reference in a self-checking GHDL testbench with random stalls on both streams (14 frames, run in CI). Board run on ZedBoard pending.

Environment: VHDL, C, TCL, Vivado, Vitis, GHDL, Zynq-7000 (xc7z020)

**EmbSIM — microcontroller and circuit simulator** personal Oct. 2026 –
github.com/EmbLinkOs/EmbSIM

- Instruction-set simulators and peripheral models (Cortex-M, AVR; STM32F1/F4 GPIO, timers, USART, SPI, I²C, ADC, DMA) coupled to a nodal-analysis circuit solver; CI runs unmodified STM32Cube, Arduino and Pico SDK examples. Early stage.

### Polytech Dijon

**Real-time VGA video pipeline on ZedBoard** Polytech, pair 2025 – 2026
github.com/teo1747/zedboard-vga-video-pipeline

- 640 × 480 chain built in IP integrator: Clocking Wizard, Video Timing Controller, AXI4-Stream video IPs, a VHDL inverter and Vitis HLS overlay IPs. Sync signals verified on hardware with ILA; the final demo showed no image because the HLS IPs had no AXI-Lite master to start them, since reworked as free-running (board retest pending).

Environment: Zynq-7000 (xc7z020), Vivado/Vitis HLS 2022.2, ILA, XDC constraints

**ZBT SRAM controller in VHDL** Polytech SoC course, pair 2025 – 2026
github.com/teo1747/vhdl-zbt-sram-controller

- FSM controller for a Micron 512K × 36 ZBT SRAM: single and 4-word burst read/write on a 36-bit bidirectional bus (Xilinx IOBUF); self-checking testbench against the vendor memory model.

Environment: VHDL, Vivado, Zynq-7000 target, Micron simulation model

**3×3 image filter on FPGA** Polytech, solo Nov. 2024 – Jan. 2025
github.com/teo1747/vhdl-2d-image-filter

- Streaming averaging and Gaussian (1-2-1/2-4-2/1-2-1) filters on 8-bit pixels: line buffers on the FIFO Generator IP, pipelined adder tree with explicit bit growth (9, 10, then 11 bits); simulated end-to-end on the Lena image, outputs compared with a software reference.

Environment: VHDL, Vivado 2024.1, Artix-7 (xc7a100t), XSim, MATLAB

**FPGA temperature monitor** Polytech, pair 2024 – 2025
github.com/teo1747/nexys4ddr-temperature-monitor

- Integrated I²C (400 kHz) and UART (115 200 baud) cores to read the on-board ADT7420 and display it on the 7-segment display; timing closed at 100 MHz (WNS +5.4 ns, 1 936 LUTs) on Nexys 4 DDR.

## Education & languages

**Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems** 2023 – Sep. 2026
Polytech Dijon Dijon, France
Graduated. Final-year apprenticeship at FORVIA. Coursework: digital design, SoC design on Zynq (Vivado, Vitis, Vitis HLS), real-time systems.

**Two-year preparatory programme — mathematics, physics, electronics** 2021 – 2023
IUC Cameroon

**Languages** English: native (anglophone), TOEIC C1 (2024) | French: fluent, professional use (B2)
**Certification** Cisco CCNA 1 (July 2024)